

Task No.: 4**Date: 08-01-2026**

Title: Develop an application that makes use of database**Aim:**

To create an Android application that uses of the databases using the SQL Databases.

Code:*MainActivity.kt*

```
package com.example.madtask5
```

```
import android.app.Activity
import android.app.AlertDialog
import android.content.Context
import android.database.Cursor
import android.database.sqlite.SQLiteDatabase
import android.os.Bundle
import android.text.Editable
import android.text.TextWatcher
import android.widget.*
```

```
class MainActivity : Activity() {
```

```
    lateinit var db: SQLiteDatabase
    private var isUpdateMode = false
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
```

```
        val etName = findViewById<EditText>(R.id.etName)
        val etRoll = findViewById<EditText>(R.id.etRoll)
        val et10 = findViewById<EditText>(R.id.et10)
        val et12 = findViewById<EditText>(R.id.et12)
```

```
        val tvAvg10 = findViewById<TextView>(R.id.tvAvg10)
```

```

val tvAvg12 = findViewById<TextView>(R.id.tvAvg12)

val spLang = findViewById<Spinner>(R.id.spLang)
val spGroup = findViewById<Spinner>(R.id.spGroup)

val btnInsert = findViewById<Button>(R.id.btnInsert)
val btnDelete = findViewById<Button>(R.id.btnDelete)
val btnUpdate = findViewById<Button>(R.id.btnUpdate)
val btnView = findViewById<Button>(R.id.btnView)
val btnViewAll = findViewById<Button>(R.id.btnViewAll)
val btnClear = findViewById<Button>(R.id.btnClear)

db = openOrCreateDatabase("StudentDB", Context.MODE_PRIVATE, null)
db.execSQL(
    "CREATE TABLE IF NOT EXISTS student(" +
        "rollno TEXT PRIMARY KEY," +
        "name TEXT," +
        "total10 REAL," +
        "avg10 REAL," +
        "language TEXT," +
        "grp TEXT," +
        "total12 REAL," +
        "avg12 REAL)"
)

spLang.adapter = ArrayAdapter(
    this,
    android.R.layout.simple_spinner_dropdown_item,
    arrayOf("Select your language", "Tamil", "French")
)

spGroup.adapter = ArrayAdapter(
    this,
    android.R.layout.simple_spinner_dropdown_item,
    arrayOf(
        "Select your group",
        "Maths Physics Chemistry Biology",
        "Maths Physics Chemistry Computer Science",
        "Commerce Economics Accountancy Business Maths",
        "Commerce Economics Accountancy Computer Application"
    )
)

```

)

```
et10.addTextChangedListener(object : TextWatcher {
    override fun afterTextChanged(s: Editable?) {
        tvAvg10.text = s.toString().toDoubleOrNull()
        ?.let { "Average of 10th: %.2f".format(it / 5) }
        ?: "Average of 10th:"
    }
    override fun beforeTextChanged(s: CharSequence?, start: Int, count: Int, after: Int) {}
    override fun onTextChanged(s: CharSequence?, start: Int, before: Int, count: Int) {}
})
```

```
et12.addTextChangedListener(object : TextWatcher {
    override fun afterTextChanged(s: Editable?) {
        tvAvg12.text = s.toString().toDoubleOrNull()
        ?.let { "Average of 12th: %.2f".format(it / 6) }
        ?: "Average of 12th:"
    }
    override fun beforeTextChanged(s: CharSequence?, start: Int, count: Int, after: Int) {}
    override fun onTextChanged(s: CharSequence?, start: Int, before: Int, count: Int) {}
})
```

```
btnInsert.setOnClickListener {
    if (etName.text.isEmpty() || etRoll.text.isEmpty()
        || spLang.selectedItemPosition == 0
        || spGroup.selectedItemPosition == 0
    ) {
        show("Error", "Please fill all details")
        return@setOnClickListener
    }
}
```

```
db.execSQL(
    "INSERT OR REPLACE INTO student VALUES(" +
        "${etRoll.text},'${etName.text}'," +
        "${et10.text},'${tvAvg10.text}'," +
        "${spLang.selectedItem},'${spGroup.selectedItem}'," +
        "${et12.text},'${tvAvg12.text}')"
)
show("Success", "Record inserted successfully")
}
```

```

btnDelete.setOnClickListener {
    if (etRoll.text.isEmpty()) {
        show("Error", "Enter Roll No")
        return@setOnClickListener
    }

    val c = db.rawQuery(
        "SELECT * FROM student WHERE rollno='${etRoll.text}'", null
    )

    if (c.moveToFirst()) {
        db.execSQL("DELETE FROM student WHERE rollno='${etRoll.text}'")
        show("Success", "Record deleted successfully")
    } else {
        show("Error", "Record not found")
    }
}

btnUpdate.setOnClickListener {
    if (etRoll.text.isEmpty()) {
        show("Error", "Enter Roll No")
        return@setOnClickListener
    }

    if (!isUpdateMode) {
        val c: Cursor =
            db.rawQuery("SELECT * FROM student WHERE rollno='${etRoll.text}'", null)

        if (c.moveToFirst()) {
            etName.setText(c.getString(1))
            et10.setText(c.getString(2))
            tvAvg10.text = c.getString(3)

            spLang.setSelection(
                (spLang.adapter as ArrayAdapter<String>)
                    .getPosition(c.getString(4))
            )

            spGroup.setSelection(
                (spGroup.adapter as ArrayAdapter<String>)
                    .getPosition(c.getString(5))
            )
        }
    }
}

```

```

    )

    et12.setText(c.getString(6))
    tvAvg12.text = c.getString(7)

    isUpdateMode = true
    show("Update", "Edit the details and press UPDATE again")
} else {
    show("Error", "Record not found")
}
} else {
    db.execSQL(
        "UPDATE student SET " +
            "name='${etName.text}'," +
            "total10='${et10.text}'," +
            "avg10='${tvAvg10.text}'," +
            "language='${spLang.selectedItem}'," +
            "grp='${spGroup.selectedItem}'," +
            "total12='${et12.text}'," +
            "avg12='${tvAvg12.text}' " +
            "WHERE rollno='${etRoll.text}'"
    )

    isUpdateMode = false
    show("Success", "Record updated successfully")
}
}

btnView.setOnClickListener {
    if (etRoll.text.isEmpty()) {
        show("Error", "Enter Roll No")
        return@setOnClickListener
    }

    val c: Cursor =
        db.rawQuery("SELECT * FROM student WHERE rollno='${etRoll.text}'", null)

    if (c.moveToFirst()) {
        val details =
            "Roll No: ${c.getString(0)}\n" +
            "Name: ${c.getString(1)}\n" +

```

```

"10th Total: ${c.getString(2)}\n" +
"10th Average: ${c.getString(3)}\n" +
"Language: ${c.getString(4)}\n" +
"Group: ${c.getString(5)}\n" +
"12th Total: ${c.getString(6)}\n" +
"12th Average: ${c.getString(7)}"

```

```

        show("Student Details", details)
    } else {
        show("Error", "Record not found")
    }
}

btnViewAll.setOnClickListener {
    val c = db.rawQuery("SELECT * FROM student", null)

    if (c.count == 0) {
        show("Error", "No records found")
        return@setOnClickListener
    }

    val buffer = StringBuilder()
    while (c.moveToNext()) {
        buffer.append("Roll No: ${c.getString(0)}\n")
        buffer.append("Name: ${c.getString(1)}\n")
        buffer.append("10th Total: ${c.getString(2)}\n")
        buffer.append("10th Average: ${c.getString(3)}\n")
        buffer.append("Language: ${c.getString(4)}\n")
        buffer.append("Group: ${c.getString(5)}\n")
        buffer.append("12th Total: ${c.getString(6)}\n")
        buffer.append("12th Average: ${c.getString(7)}\n")
        buffer.append("-----\n\n")
    }

    show("All Student Details", buffer.toString())
}

btnClear.setOnClickListener {
    etName.setText("")
    etRoll.setText("")
    et10.setText("")

```

```

        et12.setText("")
        tvAvg10.text = "Average of 10th:"
        tvAvg12.text = "Average of 12th:"
        spLang.setSelection(0)
        spGroup.setSelection(0)
        isUpdateMode = false
    }
}

private fun show(title: String, msg: String) {
    AlertDialog.Builder(this)
        .setTitle(title)
        .setMessage(msg)
        .setCancelable(true)
        .show()
}
}

activity_main.xml
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#000000">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">

        <TextView
            android:text="Name: Yoga Shri P K"
            android:textColor="#FFFFFF"
            android:textStyle="bold"
            android:textSize="18sp"
            android:layout_gravity="center_horizontal"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

```

```
<TextView
    android:text="Roll No: 23BCS192"
    android:textColor="#FFFFFF"
    android:textStyle="bold"
    android:textSize="18sp"
    android:layout_gravity="center_horizontal"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
```

```
<TextView
    android:text="STUDENT DETAILS"
    android:textColor="#FFFFFF"
    android:textStyle="bold"
    android:textSize="20sp"
    android:layout_marginTop="10dp"
    android:layout_gravity="center_horizontal"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
```

```
<EditText
    android:id="@+id/etName"
    android:hint="Enter Name"
    android:textColor="#FFFFFF"
    android:textColorHint="#B0B0B0"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<EditText
    android:id="@+id/etRoll"
    android:hint="Enter Roll No"
    android:textColor="#FFFFFF"
    android:textColorHint="#B0B0B0"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<EditText
    android:id="@+id/et10"
```

```
android:hint="Enter 10th Total Marks"  
android:inputType="number"  
android:textColor="#FFFFFF"  
android:textColorHint="#B0B0B0"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"/>
```

```
<TextView  
    android:id="@+id/tvAvg10"  
    android:text="Average of 10th:"  
    android:textColor="#FFFFFF"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<Spinner  
    android:id="@+id/spLang"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<Spinner  
    android:id="@+id/spGroup"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<EditText  
    android:id="@+id/et12"  
    android:hint="Enter 12th Total Marks"  
    android:inputType="number"  
    android:textColor="#FFFFFF"  
    android:textColorHint="#B0B0B0"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<TextView  
    android:id="@+id/tvAvg12"  
    android:text="Average of 12th:"  
    android:textColor="#FFFFFF"  
    android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"/>
```

```
<Button
```

```
    android:id="@+id/btnInsert"
    android:text="INSERT"
    android:textColor="#FFFFFF"
    android:backgroundTint="#4CAF50"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
```

```
    android:id="@+id/btnDelete"
    android:text="DELETE"
    android:textColor="#FFFFFF"
    android:backgroundTint="#F44336"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
```

```
    android:id="@+id/btnUpdate"
    android:text="UPDATE"
    android:textColor="#FFFFFF"
    android:backgroundTint="#2196F3"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
```

```
    android:id="@+id/btnView"
    android:text="VIEW"
    android:textColor="#FFFFFF"
    android:backgroundTint="#9E9E9E"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
```

```
    android:id="@+id/btnViewAll"
    android:text="VIEW ALL"
    android:textColor="#FFFFFF"
```

```
android:backgroundTint="#673AB7"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"/>
```

```
<Button
```

```
    android:id="@+id/btnClear"  
    android:text="CLEAR"  
    android:textColor="#FFFFFF"  
    android:backgroundTint="#FF9800"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```

```
</ScrollView>
```

Output:

Name: Yoga Shri P K
Roll No: 23BCS192

STUDENT DETAILS

Enter Name _____

Enter Roll No _____

Enter 10th Total Marks _____

Average of 10th: _____

Select your language ▼

Select your group ▼

Enter 12th Total Marks _____

Average of 12th: _____

INSERT

DELETE

UPDATE

VIEW

VIEW ALL

CLEAR

Name: Yoga Shri P K
Roll No: 23BCS192

STUDENT DETAILS

Yoga Shri P K

23BCS192

500

Average of 10th: 100.00

Tamil ▼

Maths Physics Chemistry Biology ▼

570

Average of 12th: 95.00

INSERT

DELETE

UPDATE

VIEW

VIEW ALL

CLEAR

Name: Yoga Shri P K
Roll No: 23BCS192

STUDENT DETAILS

Yoga Shri P K

23BCS192

500

Student Details

Roll No: 23BCS192
Name: Yoga Shri P K
10th Total: 500
10th Average: Average of 10th: 100.00
Language: Tamil
Group: Maths Physics Chemistry Biology
12th Total: 570
12th Average: Average of 12th: 95.00

UPDATE

VIEW

VIEW ALL

CLEAR

Name: Yoga Shri P K

All Student Details

Roll No: 23BCS192
Name: Yoga Shri P K
10th Total: 500
10th Average: Average of 10th: 100.00
Language: Tamil
Group: Maths Physics Chemistry Biology
12th Total: 570
12th Average: Average of 12th: 95.00

Roll No: 24MCA039
Name: Pavithra S
10th Total: 454
10th Average: Average of 10th: 90.80
Language: Tamil
Group: Maths Physics Chemistry Biology
12th Total: 478
12th Average: Average of 12th: 79.67

Roll No: 24MCA004
Name: Ancy Gratia A
10th Total: 480
10th Average: Average of 10th: 96.00
Language: Tamil

Result:

An android application using the databases are created and executed successfully.